

Apriori-Based Comparative Analysis of Cybersecurity Attack Patterns Using VCDB and UNSW-NB15

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Abstract

Cybersecurity data is collected from fundamentally different vantage points: incident-reporting repositories describe what happened to an organization, while network-traffic datasets describe how packets and sessions behave on the wire. This study asks whether Apriori-based association rule mining can reveal meaningful, interpretable attack patterns across both kinds of data, and whether the resulting patterns are comparable in nature. Apriori was applied independently to two complementary sources: the VERIS Community Database (VCDB), an incident-level repository, and UNSW-NB15, a

network intrusion-detection dataset. For VCDB, 10,596 incidents described by 66 frequency-filtered attributes (reduced from 2,683 original fields) were mined with a minimum support of 1% and a maximum itemset length of three, yielding 1,183 frequent itemsets and 5,626 candidate rules; support-confidence-lift filtering and attack-oriented restriction reduced this to 12 representative rules, headed by Downloader → System (91.01% confidence, lift 45.49) and DoS → Availability Interruption (91.12% confidence, lift 30.85). For UNSW-NB15, 175,341 training and 82,332 testing records were filtered to 161,567 and 77,910 categorical transactions respectively and mined with a minimum support of 0.005 and minimum confidence of 0.30, yielding eight representative rules validated on the held-out testing partition, headed by Protocol=unas → DoS (testing lift 9.5969) and Service=pop3 + State=FIN → Exploits (93.81% testing confidence, lift 7.5754). The two datasets were never merged into a single transaction matrix; each was mined and validated on its own terms and the results were compared afterward. The comparison shows that Apriori can surface strong, interpretable co-occurrence patterns in both incident-level and network-level data, but that the resulting support, confidence, and lift values are shaped by dataset-specific schema and preprocessing choices and are not directly comparable in magnitude. The study also confirms, using UNSW-NB15's training-testing split, that some rules generalize well while others, such as the Fuzzers pattern, do not. All discovered rules are treated strictly as statistical associations; none are interpreted as causal relationships, attack-stage sequences, or a standalone detection system.

Keywords: Cybersecurity, Association Rule Mining, Apriori Algorithm, VCDB, UNSW-NB15, Comparative Analysis, Intrusion Detection, Attack Patterns, Frequent Itemsets, Support, Confidence, Lift

I. Introduction

A. Background

The expansion of internet-connected systems and organizational digital infrastructure has increased both the volume and the diversity of cybersecurity data available for analysis. Two broad categories of such data are commonly studied. The first is incident-level data, in which analysts record what happened during a confirmed security breach — the actor involved, the action taken, the asset affected, and the resulting impact. The second is network-level data, in which raw or derived traffic features (protocol, service, connection state, and similar attributes) are logged during normal operation and during attacks. Association Rule Mining (ARM), and in particular the Apriori algorithm [1], [2], is a data-mining technique well suited to both kinds of data because it does not require a predictive target variable; instead, it surfaces items that frequently occur together and expresses this co-occurrence in the interpretable form of a rule.

B. Problem Statement

Prior work applying Apriori to cybersecurity data has generally analyzed a single dataset type in isolation — either an incident-reporting repository or a network-traffic capture — without asking whether the discovered pattern structure is a property of the analysis technique or a property of the particular dataset examined. Without a controlled comparison across dataset types, it is difficult to know how much of a given association rule's strength reflects a genuine cybersecurity pattern versus an artifact of how that specific dataset was collected and encoded.

C. Motivation for a Multi-Dataset Analysis

This study addresses that gap by applying the same underlying algorithm, evaluated with the same three metrics (support, confidence, and lift), to two datasets that represent different cybersecurity perspectives: the VERIS Community Database (VCDB), which is incident-level and describes confirmed breaches using the VERIS 4A taxonomy (Actor, Action, Asset, Attribute), and UNSW-NB15, which is network-level and describes session-level traffic characteristics with an explicit attack-category label. Analyzing both allows the study to ask not only what patterns each dataset contains, but also whether Apriori's behavior — the scale of the rules it produces, their stability, and their interpretability — is consistent across fundamentally different cybersecurity

data schemas.

D. Research Objectives

The objectives of this study are to: (1) apply Apriori independently to VCDB and to UNSW-NB15 using preprocessing appropriate to each dataset's schema; (2) generate and filter association rules for each dataset using support, confidence, and lift; (3) validate the UNSW-NB15 rules on an independent testing partition; (4) compare the resulting rule sets across a common set of criteria without merging the underlying data; and (5) interpret the comparison honestly, distinguishing statistical association from causation, prediction, and temporal attack sequencing.

E. Research Questions

The study addresses four questions. First, can Apriori association rule mining reveal meaningful, interpretable cybersecurity attack patterns in both an incident-level dataset and a network-level dataset? Second, what are the strongest associations discovered in each dataset, and how are they distributed across attack categories? Third, how do the two datasets differ in the number, scale, and stability of the rules that Apriori produces? Fourth, to what extent do the UNSW-NB15 rules generalize from training to testing data, and does this validation perspective offer any insight applicable to the VCDB analysis, which lacks a comparable train/test split?

F. Contributions of the Study

This study contributes: (i) two independent, fully documented Apriori pipelines, one for VCDB incident records and one for UNSW-NB15 network records, each preserving the dataset's native schema; (ii) a set of 12 representative VCDB rules and 8 representative UNSW-NB15 rules, evaluated with support, confidence, and lift, with the UNSW-NB15 rules additionally validated on held-out testing data; (iii) a structured, criterion-by-criterion comparative analysis of the two rule sets that avoids treating differently scaled lift and confidence values as numerically interchangeable; and (iv) an explicit discussion of what association rule mining can and cannot claim about cybersecurity attack behavior, including a clear rejection of causal, predictive, and kill-chain interpretations of the discovered rules.

II. Related Work

A. Cybersecurity Attack Pattern Analysis

Cybersecurity incident repositories such as VCDB provide structured, multi-attribute records that support quantitative analysis of attack patterns at scale. Because VCDB is built from publicly disclosed incidents using the VERIS taxonomy, it has previously been used to characterize breach trends and sector-specific risk, complementing narrower single-incident case studies. Network intrusion-detection datasets such as UNSW-NB15 serve a different analytical purpose: they support the study of how traffic-level conditions (protocol, service, connection state) relate to labelled attack categories, and they are widely used as classification benchmarks in the intrusion-detection literature.

B. Association Rule Mining in Cybersecurity

Association rule mining has been applied to several cybersecurity data sources. Li et al. [4] applied Apriori to web-page content to mine rules describing global cyberspace-security discourse. Abu et al. [5] used Apriori on Indicators of Compromise extracted from network-traffic feeds to support cyberattack attribution, evaluating rules with support, confidence, and lift. Lou et al. [6] mined association rules from multi-source security logs (IDS, firewall, and system logs) to detect intrusion behavior in cloud environments. Rahman and Williams [7] applied association rule mining to adversarial-technique co-occurrence data from the MITRE ATT&CK framework to identify techniques that adversaries tend to pair together. These studies confirm that association rule mining is an established, though comparatively under-used, approach for discovering co-occurrence structure in cybersecurity data, but each examines a single data source rather than comparing structurally different sources.

C. Apriori Algorithm

The Apriori algorithm, introduced by Agrawal, Imielinski, and Swami [3] and refined by Agrawal and Srikant [2], discovers frequent itemsets by iteratively extending frequent ($k-1$)-itemsets into candidate k -itemsets and pruning those below a minimum support threshold. Rules generated from the frequent itemsets are evaluated using support, confidence, and lift, defined formally in Section IV-I. Apriori was chosen for this study, for both datasets, because of its transparency: every retained rule can be traced directly to the underlying frequent itemset, which is important when the goal is exploratory pattern discovery rather than predictive classification.

D. VCDB-Based Cybersecurity Research

Studies using VCDB typically exploit its VERIS 4A structure (Actor, Action, Asset, Attribute) to examine which combinations of attack action, affected asset, and organizational attribute recur across the reported incident population. Because VCDB aggregates publicly disclosed breaches across many organizations and sectors, it is well suited to population-level co-occurrence analysis of the kind Apriori performs, though its incident-level granularity means it does not capture the packet- or session-level detail found in network-traffic datasets.

E. UNSW-NB15-Based Cybersecurity Research

UNSW-NB15 [9] was developed at UNSW Canberra to provide a modern benchmark combining normal traffic with nine synthetic contemporary attack categories, distributed as separate training and testing CSV partitions. It has been used extensively for supervised intrusion-detection classification; its use for unsupervised association rule mining, and specifically for training/testing validation of the discovered rules rather than of a trained classifier, is comparatively less common.

F. Research Gap

A substantial portion of cybersecurity data-mining research concentrates on classification, anomaly detection, and intrusion prediction using machine-learning and deep-learning models [3]. These approaches are optimized to detect or predict individual attacks rather than to describe how attack characteristics, assets, and

impacts co-occur across a broad population, and they are almost always applied to a single dataset. This is not a claim that no prior study has used Apriori in a cybersecurity context, as Section II-B shows several have. Rather, the gap addressed here is the controlled comparison of Apriori's behavior across two structurally different cybersecurity data sources — incident-level (VCDB) and network-level (UNSW-NB15) — analyzed separately and compared afterward, rather than the application of Apriori to either dataset in isolation.

III. Datasets

A. VERIS Community Database (VCDB)

VCDB [8] is a publicly available repository of security incidents structured using the VERIS 4A framework (Actor, Action, Asset, Attribute). The working extract used in this study contains 10,596 incident records described by 2,683 original fields. Because the raw dataset is highly dimensional, a frequency-filtered subset of 66 attributes, drawn from ten VERIS attribute groups (hacking, malware, social-engineering, misuse, and error actions; asset varieties; external-actor varieties; and confidentiality, integrity, and availability impact varieties), was retained for association analysis.

B. UNSW-NB15

UNSW-NB15 [9] contains modern normal activity together with nine synthetic contemporary attack categories (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms). The official dataset description reports 2,540,044 total records and 49 generated features with a class label; the supplied training partition contains 175,341 records and the testing partition contains 82,332 records. Four categorical fields — proto, service, state, and attack_cat — were selected for the transaction representation used in this study.

C. Comparative Characteristics of the Datasets

Table I summarizes the two datasets against a common set of criteria. The comparison highlights that VCDB and UNSW-NB15 differ not only in scale but in what a single transaction represents: a VCDB transaction is a confirmed, human-investigated incident, while a UNSW-NB15 transaction is a categorical summary of a single network session labelled by an automated or synthetic process.

Criterion	VCDB	UNSW-NB15
Data perspective	Incident-level (post-hoc breach reporting)	Network-level (session traffic)
Records used	10,596 incidents	175,341 train / 82,332 test
Original features	2,683 fields	49 features + class label
Features after filtering	66 attributes	4 categorical fields (proto, service, state, attack_cat)
Attack representation	VERIS 4A: Actor, Action, Asset, Attribute	9 labelled attack categories
Transaction unit	One incident = one binary transaction (66 items)	One session = one categorical transaction
Train/test availability	Single partition (no official split)	Official train/test CSV partitions

Criterion	VCDB	UNSW-NB15
Purpose in this study	Incident-level co-occurrence discovery	Network-condition association + generalization check

Table I. Comparative characteristics of VCDB and UNSW-NB15.

IV. Proposed Methodology

A. Overall Research Framework

The two datasets were never combined into a single transaction matrix; their schemas are not compatible and merging them would obscure rather than clarify each data source's own structure. Instead, each dataset was preprocessed, transformed into transactions, and mined with Apriori using thresholds appropriate to its own scale and sparsity, exactly as in the two source studies. The resulting rule sets were compared only after each had been independently generated and, for UNSW-NB15, independently validated. Fig. 1 illustrates this analyze-separately-then-compare workflow.

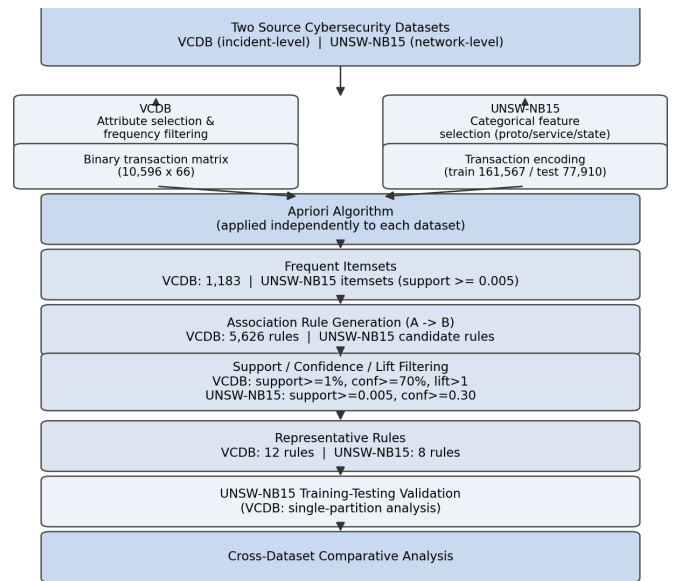


Fig. 1. Overall proposed research methodology. VCDB and UNSW-NB15 are preprocessed, transformed, and mined independently; only the resulting rule sets are compared.

B. VCDB Data Preprocessing

Fields related to attack actions, malware activity, social engineering, misuse, errors, affected assets, external actors, and confidentiality/integrity/availability impacts were considered; country-level, notes, and unknown-category fields were excluded. Retained fields were converted into a binary representation (1 = present, 0 = absent). Attributes with very low occurrence were removed using frequency-based filtering: attack/action attributes were retained from a 0.5% minimum occurrence threshold, while contextual and impact attributes were retained from a 1% threshold. After filtering, 66 attributes remained across the 10,596 incidents.

C. VCDB Attribute Selection

Candidate fields were drawn from ten VERIS attribute groups: hacking actions, malware actions, social-engineering actions, misuse actions, error actions, asset varieties, external-actor varieties, and confidentiality, integrity, and availability impact varieties. Applying the occurrence thresholds above to these

groups yielded the final set of 66 attributes used to construct the transaction matrix.

D. VCDB Transaction Representation

Each incident was treated as one transaction, and each of the 66 selected attributes as one item; a value of 1 indicates the item was present in that incident. For example, an incident involving phishing that affected a person-type asset would be represented as the transaction {Phishing, Person}. This produced a $10,596 \times 66$ binary transaction matrix suitable as input to Apriori.

E. UNSW-NB15 Data Preprocessing

The Apriori algorithm operates on item-based transactions rather than raw continuous measurements, so four categorical fields (proto, service, state, attack_cat) were selected for the transaction representation. To reduce very sparse categories, commonly occurring values were retained: protocol values tcp, udp, unas, arp, ospf, and sctp; service values -, dns, http, smtp, ftp-data, ftp, ssh, and pop3; and state values INT, FIN, CON, and REQ. After this filtering step, 161,567 of the 175,341 training records remained, and the same procedure applied to the testing data left 77,910 records for validation.

F. UNSW-NB15 Feature Selection

Unlike VCDB, where 66 attributes were retained from a large pool of incident fields, UNSW-NB15's transaction representation deliberately uses a small number of categorical network-condition fields. This reflects the difference in what each dataset is designed to describe: VCDB attributes span actors, actions, assets, and impacts, whereas the UNSW-NB15 fields used here describe only the network-session conditions under which a given attack category was recorded.

G. UNSW-NB15 Transaction Representation

Each retained record was converted into a transaction containing items such as Protocol=tcp, Service=dns, and State=INT, together with the corresponding attack category. The same categorical filtering procedure was applied to both the training and testing partitions, so that the testing transactions were constructed identically to the training transactions before validation.

H. Apriori Association Rule Mining

For VCDB, Apriori was applied to identify frequent one-, two-, and three-itemsets among the 66 items, using a minimum support threshold of 1% and a maximum itemset length of three, producing 1,183 frequent itemsets (54 one-itemsets, 287 two-itemsets, and 842 three-itemsets) and, from these, 5,626 association rules. For UNSW-NB15, frequent itemsets were generated with a minimum support of 0.005 (0.5%), and association rules were generated with a minimum confidence of 0.30 (30%); candidate rules were restricted to those whose consequent contained an attack category, while attack items were excluded from antecedents, and rules predicting the normal class were excluded from the final analysis.

I. Support, Confidence, and Lift

For an association rule $A \rightarrow B$, support measures the proportion of transactions containing both A and B; confidence measures the proportion of transactions containing A that also contain B ($\text{Support}(A \rightarrow B) / \text{Support}(A)$); and lift compares the observed confidence to the confidence expected under independence ($\text{Confidence}(A \rightarrow B) / \text{Support}(B)$). A lift greater than 1 indicates positive association, a value near 1 indicates near-independence, and a value below 1 indicates negative association. Both datasets were evaluated using these same three definitions, applied to each dataset's own transaction matrix; the numeric thresholds used for filtering differ between datasets (Section IV-J) because the two matrices differ substantially in size and sparsity.

J. Rule Filtering and Selection

VCDB rules were filtered using support $\geq 1\%$, confidence $\geq 70\%$, and lift > 1 , yielding 2,744 meaningful rules; restricting to rules whose antecedent represented an attack action produced 1,901 attack-oriented rules, and restricting further to simple one-to-one attack $\rightarrow$ impact/context rules produced 76 candidates, from which 12 representative rules (one per attack pattern of interest) were selected for detailed analysis. UNSW-NB15 candidate rules were filtered to those meeting the 0.005 support and 0.30 confidence thresholds, with attack categories restricted to the consequent; redundant variants with essentially equivalent antecedent information were removed to obtain eight representative rules covering the DoS, Exploits, Generic, and Fuzzers categories.

K. UNSW-NB15 Training-Testing Validation

Each of the eight UNSW-NB15 representative rules was evaluated a second time on the independently filtered testing partition (77,910 transactions), using the same antecedent/consequent definitions derived from training. Testing confidence and testing lift were then compared against their training counterparts. VCDB was not analyzed this way because the working extract used in this study does not provide an official training/testing split; its 12 representative rules therefore reflect analysis of a single partition and should be read with that difference in mind when compared to the UNSW-NB15 results.

L. Comparative Analysis Framework

The comparison in Section VII is organized around twelve criteria: nature of the underlying cybersecurity data, incident-level versus network-level perspective, transaction construction, attack representation, number and type of rules produced, strongest associations found, support, confidence, lift, availability of generalization/validation evidence, interpretability, and the practical strengths and limitations of applying Apriori to each data type. Because VCDB and UNSW-NB15 differ in transaction count, item count, and filtering thresholds, the comparison discusses lift and confidence values in their dataset-specific context rather than treating them as directly interchangeable numbers.

V. VCDB Results and Discussion

A. Overall Apriori Results

Applying Apriori to the $10,596 \times 66$ transaction matrix with a minimum support of 1% and a maximum itemset length of three produced 1,183 frequent itemsets (54 one-itemsets, 287 two-itemsets, 842 three-itemsets). Rule generation from these itemsets produced 5,626 association rules; support/confidence/lift filtering reduced this to 2,744 meaningful rules, of which 1,901 were attack-oriented. After restricting to one-to-one attack $\rightarrow$ impact/context rules and selecting a representative rule per attack pattern of interest, 12 rules were retained for detailed discussion.

B. Selected Association Rules

Table II lists the 12 selected rules together with their incident counts, support, confidence, and lift.

Category	Pattern	Outcome	N	Supp. %	Conf. %	Lift
Soc. Eng.	Phishing	Alter Behavior	450	4.25	100.00	15.05
Soc. Eng.	Phishing	Person	450	4.25	100.00	14.30
Hacking	DoS	Availability Interrupt.	154	1.45	91.12	30.85
Hacking	Exploit Vuln.	Server	1042	9.83	97.75	1.75
Malware	Ransomware	Server	1133	10.69	94.73	1.69
Misuse	Privilege Abuse	Server	980	9.25	79.48	1.42
Error	Loss	Availability Loss	420	3.96	100.00	5.09
Error	Disposal Error	Media	298	2.81	90.58	4.15
Error	Misconfiguration	Server	373	3.52	96.13	1.72
Malware	Spyware/Keylogger	Credentials	116	1.09	77.85	10.32
Malware	Downloader	System	162	1.53	91.01	45.49
Malware	C2	System	164	1.55	80.39	40.18

Table II. VCDB selected association rules ($n = 10,596$ incidents).

Malware-related rules produced the highest lift values: Downloader $\rightarrow$ System reached a lift of 45.49 with 91.01% confidence, and C2 $\rightarrow$ System reached 40.18 with 80.39% confidence. The hacking-related rule DoS $\rightarrow$ Availability Interruption showed a lift of 30.85 and confidence of 91.12%. Both selected social-engineering rules, Phishing $\rightarrow$ Alter Behavior and Phishing $\rightarrow$ Person, achieved 100% confidence, with lift values of 15.05 and 14.30 respectively. In contrast, Exploit Vulnerability $\rightarrow$ Server combined comparatively high support (9.83%) with 97.75% confidence but a lower lift of 1.75, illustrating that support, confidence, and lift capture different, complementary aspects of a rule and should be read jointly.

C. Category-Wise Analysis

Attack Category	No. of Rules	Share (%)
Malware	4	33.33
Error	3	25.00

Attack Category	No. of Rules	Share (%)
Social Engineering	2	16.67
Hacking	2	16.67
Misuse	1	8.33
Total	12	100.00

Table III. Category-wise distribution of the 12 selected VCDB rules.

Malware accounts for the largest share of selected rules (33.33%), covering Ransomware, Spyware/Keylogger, Downloader, and C2. Error-related patterns represent 25.00% of the set, showing that non-malicious incidents also produce strong asset and availability associations. Social Engineering and Hacking each contribute 16.67%, and Misuse contributes 8.33% (Privilege Abuse $\rightarrow$ Server). The distribution indicates that the strongest associations are not confined to a single attack category.

D. Support, Confidence and Lift Analysis

Support indicates prevalence across the 10,596 incidents: Ransomware $\rightarrow$ Server has the highest support (10.69%), followed by Exploit Vulnerability $\rightarrow$ Server (9.83%), while Spyware/Keylogger $\rightarrow$ Credentials, Downloader $\rightarrow$ System, and C2 $\rightarrow$ System have lower support (1–2%) despite satisfying the minimum threshold. Confidence indicates consistency: Phishing $\rightarrow$ Alter Behavior, Phishing $\rightarrow$ Person, and Loss $\rightarrow$ Availability Loss each reach 100% confidence. Lift is highest for Downloader $\rightarrow$ System (45.49), C2 $\rightarrow$ System (40.18), and DoS $\rightarrow$ Availability Interruption (30.85). High lift for comparatively rare consequents should be read carefully, since rare attributes can mechanically produce large lift values.

E. Strongest Associations

The strongest lift does not necessarily correspond to the most prevalent pattern: Downloader $\rightarrow$ System has the highest lift but a comparatively low support of 1.53%, whereas Ransomware $\rightarrow$ Server has the highest support (10.69%) but a much lower lift (1.69). Malware-related attributes produced the highest-lift associations overall, while hacking (DoS) and social-engineering (Phishing) rules also showed strong, high-confidence associations, and non-malicious error-related incidents produced associations comparable in strength to several deliberate-attack categories.

F. Interpretation and Findings

These associations describe statistical co-occurrence within the VCDB records; they do not establish causation, and because Apriori does not encode event ordering, the rules do not represent temporal attack sequences or kill-chain stages. Fig. 2 compares the lift of the 12 selected rules, confirming Downloader $\rightarrow$ System, C2 $\rightarrow$ System, and DoS $\rightarrow$ Availability Interruption as the three highest-lift rules.

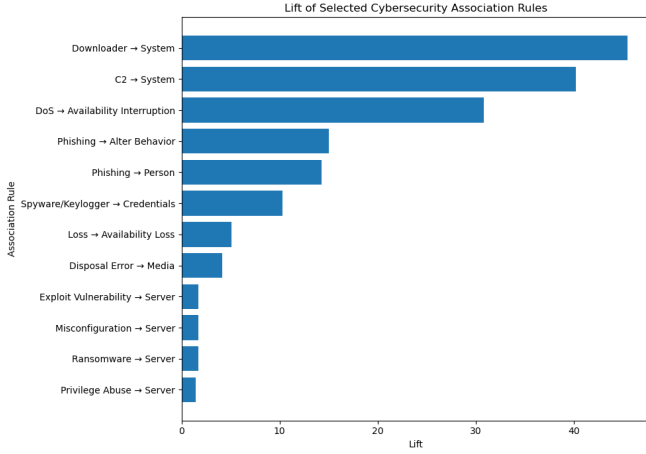


Fig. 2. VCDB selected-rule lift comparison.

Fig. 3 plots support against confidence for the 12 rules; Ransomware → Server and Exploit Vulnerability → Server combine relatively high support with high confidence, while several malware-related rules show high confidence at comparatively low support.

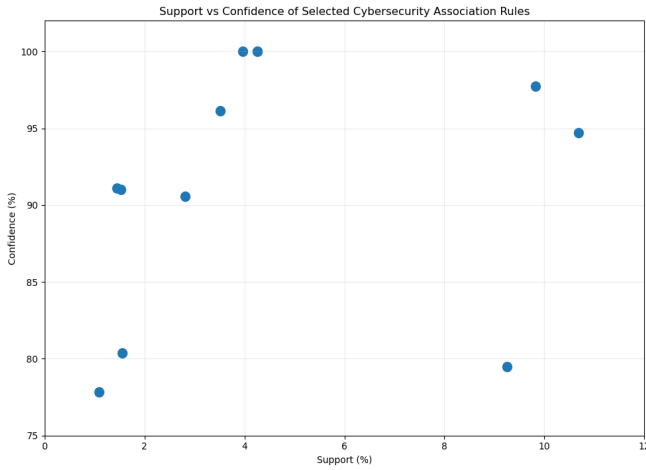


Fig. 3. VCDB support vs. confidence of selected rules.

Fig. 4 shows the category-wise distribution of the 12 selected rules, corresponding to Table III.

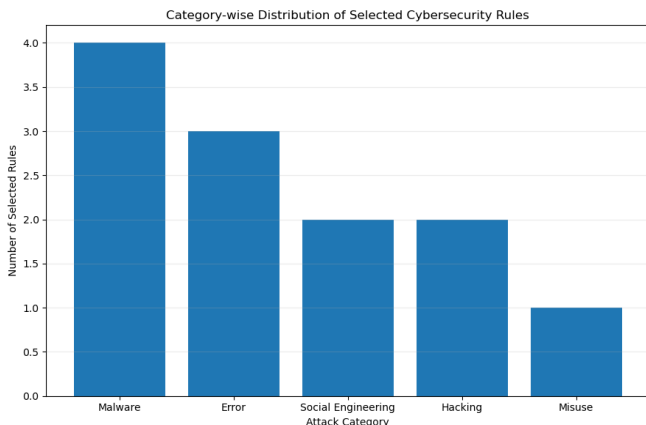


Fig. 4. Category-wise distribution of selected VCDB rules.

VI. UNSW-NB15 Results and Discussion

A. Final Representative Rules

Table IV lists the eight representative UNSW-NB15 rules with their training and testing confidence and lift.

Attack Pattern	Train Conf.	Test Conf.	Train Lift	Test Lift
DoS: Protocol=unas	33.76%	33.17%	7.0772	9.5969
Exploits: pop3+FIN	94.27%	93.81%	5.5809	7.5754
Exploits: smtp	61.54%	56.67%	3.6430	4.5764
Generic: dns+INT	99.20%	99.44%	4.0226	4.1147
Generic: udp+dns	82.78%	85.06%	3.3568	3.5194
Exploits: ospf+INT	49.60%	41.22%	2.9366	3.3287
Generic: udp+INT	76.17%	74.21%	3.0887	3.0708
Fuzzers: udp+-+INT	37.48%	16.77%	3.4583	2.4095

Table IV. UNSW-NB15 representative association rules and training/testing metrics.

B. DoS Association Pattern

The strongest relative association was Protocol=unas → DoS. Training confidence was 33.76% and testing confidence was 33.17%, a change of only −0.59 percentage points, while lift increased from 7.0772 in training to 9.5969 in testing. The protocol condition remained strongly associated with DoS relative to DoS prevalence in the filtered testing data, though confidence remains substantially below 100%, so not every record using this protocol is a DoS attack.

C. Exploits Association Patterns

The strongest high-confidence rule was Service=pop3 + State=FIN → Exploits (94.27% training, 93.81% testing confidence; lift rising from 5.5809 to 7.5754), showing strong consistency across partitions. Service=smtp → Exploits decreased in confidence from 61.54% to 56.67% while its lift increased from 3.6430 to 4.5764. Protocol=ospf + State=INT → Exploits decreased from 49.60% to 41.22% confidence but retained a testing lift of 3.3287.

D. Generic Association Patterns

Service=dns + State=INT → Generic achieved 99.20% training and 99.44% testing confidence, with lift rising from 4.0226 to 4.1147. Protocol=udp + Service=dns → Generic generalized well (testing confidence 85.06%, testing lift 3.5194), and Protocol=udp + State=INT → Generic achieved 74.21% testing confidence and 3.0708 testing lift, together indicating consistent associations involving DNS, UDP, and the INT state.

E. Fuzzers Pattern

The representative Fuzzers rule, Protocol=udp + Service=- + State=INT → Fuzzers, showed the largest reduction in confidence, from 37.48% in training to 16.77% in testing, while lift decreased from 3.4583 to 2.4095. Although testing lift remained above 1, the substantial change indicates weaker generalization than the strongest rules, and demonstrates that a rule satisfying the mining thresholds on training data may still behave differently on unseen data.

F. Training-Testing Comparison

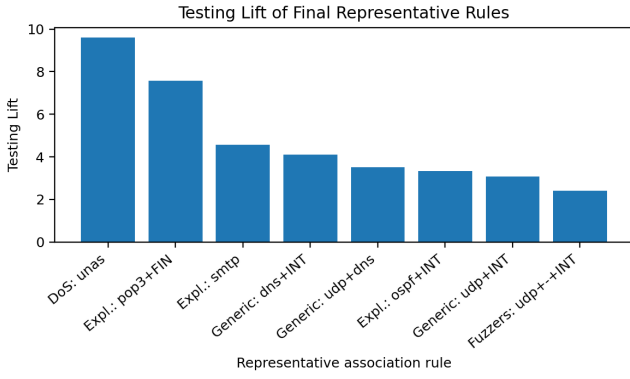


Fig. 5. UNSW-NB15 testing lift for the eight representative rules.

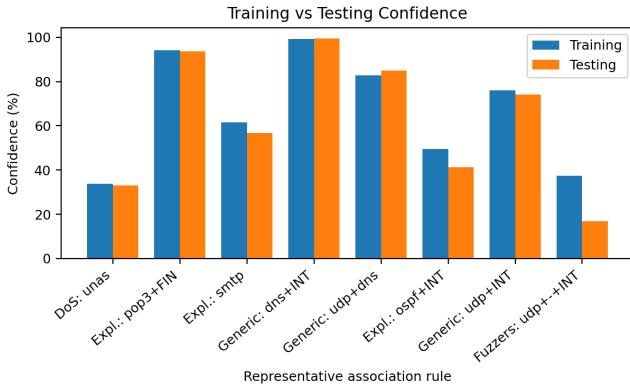


Fig. 6. UNSW-NB15 training vs. testing confidence.

Across the eight rules, the POP3–FIN Exploits and DNS–INT Generic associations show the closest training/testing agreement, the DoS rule shows a small confidence change while its relative lift increases in testing, and the Fuzzers rule is the main exception, with a pronounced decline in both confidence and lift.

G. Interpretation of Lift and Confidence

The DoS rule has the highest testing lift (9.5969) but only 33.17% testing confidence, while the POP3–FIN Exploits rule has 93.81% testing confidence and a lower, though still substantial, testing lift of 7.5754. High lift indicates that an event is much more common under the antecedent than expected from its overall prevalence, whereas confidence describes how often the consequent occurs given the antecedent; the two measures answer different questions and should not be conflated.

VII. Comparative Analysis

This section compares the VCDB and UNSW-NB15 results directly, criterion by criterion, while keeping in view that the two datasets differ in transaction count, item count, and the meaning of a transaction.

Criterion	VCDB	UNSW-NB15
Min. support / confidence used	1% / 70%	0.5% (0.005) / 30% (0.30)
Max itemset length	3	Not separately restricted
Frequent itemsets	1,183 (54/287/842)	Not reported as a single total; filtered by rule stage
Candidate rules generated	5,626	Filtered directly to attack-only, non-normal candidates

Criterion	VCDB	UNSW-NB15
Meaningful / attack-oriented rules	2,744 meaningful; 1,901 attack-oriented	N/A (filtering criteria differ; see IV-J)
Representative rules retained	12	8
Highest lift (own scale)	45.49 (Downloader to System)	9.5969 (Protocol=unas to DoS, testing)
Highest confidence	100% (two Phishing rules; Loss rule)	99.44% (Service=dns+State=INT to Generic, testing)
Validation approach	Single-partition analysis (no official split)	Independent training/testing validation

Table V. Comparative Apriori results for VCDB and UNSW-NB15.

VCDB produced considerably higher raw lift values than UNSW-NB15 (up to 45.49 versus 9.5969). This difference should not be read as VCDB containing "stronger" cybersecurity patterns; lift is sensitive to the base rate of the consequent, and VCDB's rare, specific outcome items (such as "System" as a consequent of Downloader) can mechanically produce larger lift values than UNSW-NB15's attack-category consequents, which are comparatively less rare within their own filtered transaction set. The two lift scales are therefore not directly comparable in magnitude; they are comparable only in the qualitative sense that both exceed 1 and indicate genuine positive association within each dataset's own transaction population.

VCDB produced very high-lift associations such as Downloader → System and C2 → System, reflecting the incident-level view in which a specific malware action is strongly tied to a specific affected system type. UNSW-NB15 produced strong network-condition associations such as Protocol=unas → DoS, reflecting the network-level view in which traffic conditions co-occur with a labelled attack category. VCDB captures broader incident characteristics — actors, actions, assets, and organizational impacts — while UNSW-NB15 captures narrower network traffic characteristics. UNSW-NB15 additionally provides a training/testing validation perspective that VCDB, in the form analyzed here, does not: some UNSW-NB15 associations (POP3–FIN Exploits, DNS–INT Generic) are highly stable between training and testing, while others, such as the Fuzzers pattern, generalize considerably less well.

In terms of interpretability, both rule sets remain human-readable: a VCDB rule such as Phishing → Alter Behavior and a UNSW-NB15 rule such as Service=pop3 + State=FIN → Exploits are both expressed as simple antecedent-consequent statements that a non-specialist analyst can read directly. The practical strength of Apriori on VCDB is its ability to surface cross-category patterns (malware, hacking, social engineering, misuse, error) from a very high-dimensional incident schema; its limitation is the absence of an independent validation partition. The practical strength of Apriori on UNSW-NB15 is the availability of that validation partition, which exposes rules such as the Fuzzers pattern as unstable; its limitation is that the categorical feature set used here is comparatively narrow (protocol, service, state) and does not draw on UNSW-NB15's many numerical traffic features.

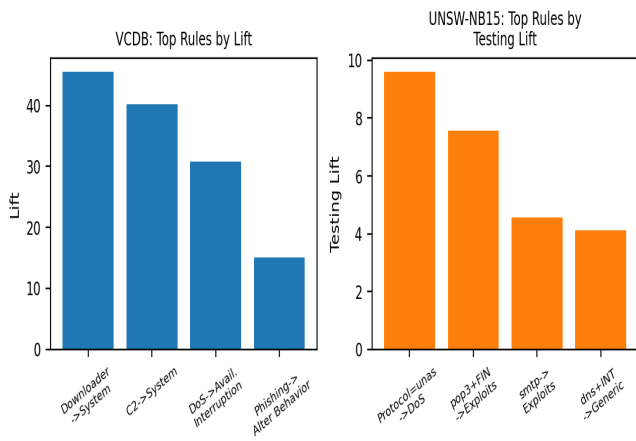


Fig. 7. Comparative visualization of the top rules by lift in each dataset, shown on separate axes because the two lift scales are not directly comparable in magnitude.

VIII. Overall Discussion

A. Major Findings

Apriori discovered interpretable, non-trivial associations in both datasets: malware-, hacking-, and social-engineering-related rules in VCDB, and protocol/service/state-based rules in UNSW-NB15. In both cases, the strongest lift did not necessarily correspond to the highest support or confidence, confirming that the three metrics measure distinct properties of a rule and must be read together.

B. Cross-Dataset Insights

The comparison shows that Apriori's behavior is consistent in kind but not in scale across dataset types: both datasets yield a small set of high-value representative rules after aggressive filtering, but the number of candidate rules, the filtering thresholds required, and the resulting lift magnitudes all depend heavily on the dataset's schema, dimensionality, and transaction definition.

C. Practical Interpretability

Both rule sets remain usable by a human analyst without additional modeling: each rule states, in plain terms, which conditions or actions were disproportionately associated with which outcome or attack category in the analyzed data.

D. Implications for Cybersecurity Analytics

For incident-response teams working with VCDB-like data, the discovered associations (e.g., Downloader → System, DoS → Availability Interruption) can help prioritize which asset types warrant closer monitoring given a detected action. For network-operations teams working with UNSW-NB15-like data, the discovered associations (e.g., Protocol=unas → DoS) can flag network conditions worth deeper inspection, with the important caveat, demonstrated directly by the Fuzzers result, that any such rule should be validated on independent data before being trusted.

E. Why Association Mining Complements Classification-Based Methods

Association rule mining does not require a labelled target and produces directly interpretable output, in contrast to most classification and deep-learning intrusion-detection methods, which optimize predictive accuracy but do not by themselves explain which conditions are associated with which outcomes. The rules produced here are best viewed as a complementary exploratory layer that can motivate features or hypotheses for a subsequent classification model, not as a replacement for one; this study does not claim, and the evidence does not support, that these rules constitute a standalone intrusion detection system.

IX. Limitations

A. Dataset Bias

VCDB is built from publicly reported incidents and may not represent the full population of security incidents. UNSW-NB15 is a synthetic testbed capture and may not represent every real-world network environment. Neither dataset's rules should be assumed to generalize automatically to a different organizational or network context.

B. Categorical Representation

Both pipelines represent transactions using categorical or binary items. UNSW-NB15's transaction representation focuses on protocol, service, and connection state and does not directly represent continuous network characteristics, so information in numerical traffic features may be missed.

C. Attribute/Feature Selection

VCDB's attribute selection and frequency-based filtering, necessary given the original 2,683-field dataset, excludes rarer characteristics from the final analysis. UNSW-NB15's feature selection is restricted to four categorical fields, so the discovered rules describe only a narrow slice of the dataset's available information.

D. Threshold Sensitivity

Different minimum support and confidence thresholds were required for each dataset (1%/70% for VCDB, 0.5%/30% for UNSW-NB15) because of differences in transaction count and item sparsity. Different thresholds in either dataset can produce different frequent itemsets and rules, and patterns below the chosen support threshold are not represented in either analysis.

E. Rare Attack Categories

In VCDB, rarer attack patterns beyond the 12 selected rules were not analyzed in depth. In UNSW-NB15, the final representative rules are concentrated on DoS, Exploits, Generic, and Fuzzers; less frequent categories did not produce representative rules under the selected criteria.

F. Dataset-Specific Behaviour

The UNSW-NB15 findings are based on its official testing partition, which provides within-dataset validation but not evaluation on an independent intrusion-detection dataset. The VCDB findings are based on a single working extract with no comparable external validation set.

G. Lack of Causality

The rules from both datasets are descriptive statistical associations. They do not establish causal relationships and should not be presented as a standalone predictive classifier for either dataset.

H. Lack of Temporal Ordering

Each VCDB incident and each UNSW-NB15 session is treated as a single, unordered transaction. Neither analysis encodes chronological ordering, so the rules should not be read as attack sequences or kill-chain stages in either dataset.

I. Limited Itemset Length

VCDB itemsets were restricted to a maximum length of three attributes, so larger multi-attribute relationships were not explored. UNSW-NB15's representative rules similarly combine at most three categorical conditions.

J. Apriori Computational Scalability

Apriori can become computationally demanding as the number of items and transaction complexity increase. Larger or higher-dimensional cybersecurity datasets, in either the incident-level or network-level category, may require pruning, optimization, or alternative algorithms such as FP-Growth.

K. Need for Independent External Datasets and UNSW-NB15 Generalization Differences

The Fuzzers rule's large confidence decline between training and testing shows that some discovered associations are sensitive to data distribution even within a single dataset; broader generalization of either dataset's rules requires evaluation on independent, external cybersecurity data, which is left to future work.

X. Conclusion

This study applied Apriori-based association rule mining independently to two datasets representing different cybersecurity perspectives — the incident-level VERIS Community Database (VCDB) and the network-level UNSW-NB15 intrusion-detection dataset — in order to ask whether meaningful, interpretable attack patterns can be recovered from both, and whether the resulting patterns are comparable. Two datasets were used, rather than one, because a single dataset type cannot show whether a discovered association reflects a genuine cybersecurity pattern or an artifact of that dataset's own schema and collection process.

From 10,596 VCDB incidents and 66 selected attributes, Apriori identified 1,183 frequent itemsets and 5,626 candidate rules, narrowed after filtering to 12 representative rules headed by Downloader → System (lift 45.49), C2 → System (lift 40.18), and DoS → Availability Interruption (lift 30.85). From 175,341 UNSW-NB15 training records and 82,332 testing records, filtered to 161,567 and 77,910 categorical transactions respectively, Apriori produced eight representative rules headed, on the testing partition, by Protocol=unas → DoS (lift 9.5969) and Service=pop3 + State=FIN → Exploits (93.81% confidence, lift 7.5754).

Support, confidence, and lift were essential throughout, since no single measure was sufficient on its own: several of the strongest-lift rules in both datasets had comparatively low support, while several high-support rules had comparatively low lift. The UNSW-NB15 training/testing validation further showed that rule strength is not automatically stable — most representative rules generalized closely from training to testing, but the Fuzzers pattern's confidence fell from 37.48% to 16.77%, demonstrating why independent validation matters whenever it is available.

The comparative analysis found that Apriori behaves consistently in kind, but not in numerical scale, across the two dataset types: both produce a small set of interpretable, high-value rules after aggressive filtering,

but the raw lift and confidence values are shaped by each dataset's own transaction count, item sparsity, and schema, and are not directly comparable in magnitude between VCDB and UNSW-NB15. The overall contribution of this study is therefore not a single best rule set, but a demonstration that Apriori-based association rule mining is a viable, complementary, and honestly interpretable exploratory tool across structurally different cybersecurity data sources, provided its output is read as statistical association rather than causation, prediction, or attack sequencing, and provided its limitations — dataset bias, threshold sensitivity, limited itemset length, and (where unavailable) the absence of independent validation — are kept in view.

XI. Future Scope

11.1 Compare Apriori against FP-Growth and Eclat on both datasets in terms of computational efficiency, pattern quality, and interpretability.

11.2 Explore itemsets larger than three attributes in both VCDB and UNSW-NB15 to capture richer multi-attribute relationships.

11.3 Incorporate discretization or hybrid transaction-generation techniques so that UNSW-NB15's numerical traffic features can be included alongside the categorical fields used here.

11.4 Extend the comparative approach to additional cybersecurity datasets beyond VCDB and UNSW-NB15 to assess how consistently Apriori's behavior holds across even more data sources.

11.5 Validate VCDB's representative rules against an independent incident dataset, giving VCDB the same kind of generalization check that UNSW-NB15's training/testing split already provides.

11.6 Investigate incremental or streaming association-rule mining so that both incident-reporting and network-traffic pipelines can be updated as new data arrives, rather than re-mined from scratch.

11.7 Combine association mining with supervised classification methods in a hybrid framework, using discovered rules as engineered features or as candidate hypotheses for a downstream classifier.

11.8 Develop interactive cybersecurity analytics tools that let analysts explore VCDB- and UNSW-NB15-derived associations side by side, with the dataset-specific caveats from Section VII built into the interface.

11.9 Develop automated rule-selection and redundancy-reduction methods to keep the retained rule set compact and informative as either dataset grows.

11.10 Apply sequence-mining techniques to study the order and progression of events, which would address temporal attack sequencing in a way this co-occurrence-based study explicitly does not attempt.

XII. References

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Research-data note. All VCDB numerical results (incident counts, itemset counts, rule counts, and the support/confidence/lift values in Tables II and III) are reproduced unchanged from the VCDB source study, and all UNSW-NB15 numerical results (record counts and the support/confidence/lift values in Table IV) are reproduced unchanged from the UNSW-NB15 source study. No numerical result in this paper was invented, estimated, or adjusted; where a comparison required a new figure (Fig. 1 and Fig. 7), it was constructed only from these same reported values.